



IRAMCO INSTITUTE OF TECHNOLOGY
RAJAPALAYAM, TAMILNADU, INDIA

DEPT. OF COMPUTER SCIENCE AND ENGINEERING
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Course code & Title: IT6801-Service Oriented Architecture

Semester - VII Date: 25.10.2019, Time: 4.10 pm to 500 pm

Topic: Web Service and its Applications

FLIPPED CLASS COLLABORATIVE LEARNING

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4. Topic: Web service and its applications

4.1 Objectives

- To enhance communication, listening, Self-confidence and problem solving skills
- To describe the Service Oriented Principles.
- To apply Service Oriented Principles to develop a business automation system

4.2 Outcome (O) (At the end of the learning technique, the students will be able to)

- Describe the Service Oriented principles to develop web service based applications.

4.3 Justification for choosing the topic

Service-oriented Principles SOP geared towards recent advancement in industries/organization. SOP are core parts of my course namely, Service-Oriented Architecture. SOP enables software integration with legacy systems. The prime industries like Oracle, IBM, Microsoft, and Infosys endeavor to enhance their legacy systems with SOP to develop an automation process. The objective of collaborative learning is to enhance communication, creativity and problem-solving skills. This domain helps the students to get placement in core fields.

A collaborative learning technique consists of a series of assessments that enable students to master the given concept/topic. It emphasizes effective group interaction among students to master particular concepts. Also, this learning technique enhances the students with communication, and problem-solving skills. It is not more attention in part of the syllabus and examination because the textbook contains only five pages. So, I consider this topic is *independent study*. Since all the students like autonomy and want to learn their own

4.4 Flipped classroom

Flipped classroom is a pedagogical approach [1] in which traditional content delivery moves to an *individual learning process*, and the group activity is transformed into a dynamic, interactive learning environment where the instructor just guides the learners to apply their creative thinking to provide a fruitful solution in the course content. It provides more benefits than traditional direct content delivery.

4.4.1 Benefits of flipped classroom

- It promotes peer interaction among students and *collaboration skills*
- It encourages bright students' engagement.
- It provides increased individual attention.
- It makes *independent learning* rather than teaching.

4.4.2 Procedure:



Figure 1: Steps of Flipped Classroom

4.4.2.1 Plan

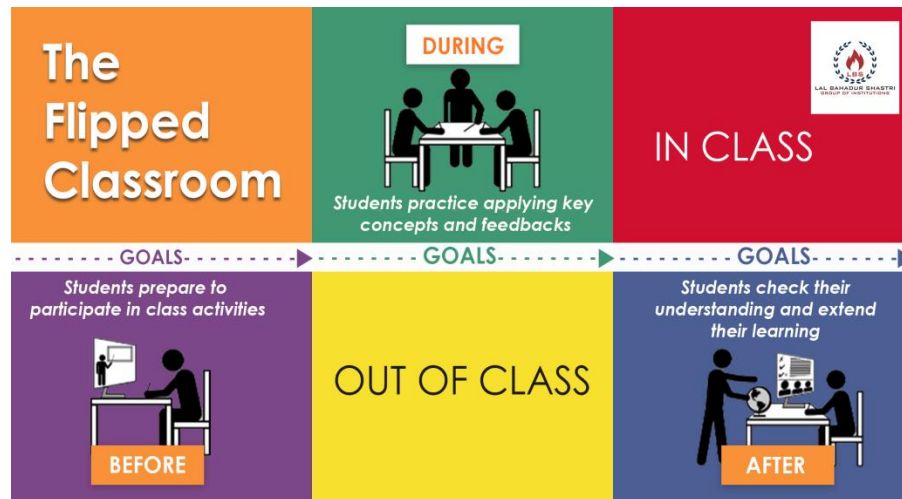
- Identify the topic for *independent study*.
- Write the *objectives* of the flipped class
- Create the *schedules* for a flipped class that contains the dates for posting material, class activity, and quiz. – Select the dates before holiday or exam.

4.4.2.2 Prepare

- Create an assessment (formative/summative) for each activity
- Identify existing recorded video from MOOCs (NPTEL/MIT etc.) and text materials or create a video lecture or
- Write the guidelines for learning from the video

4.4.2.3 Implement

- Form a small group. A group may be heterogeneous or homogeneous based on the students' performance.
- Upload the learning materials such as video and text documents on the course website
- Give enough time(minimum 3 to 5 days) for students to study
- Post a quiz to find whether the students understood the study materials or not and address the gaps before in class activity.
- Design an application-oriented problem or project or case study for a class activity
- Monitor the class activity to ensure that each group is actively involved to provide the solutions.
- Make the group presentation for providing solutions.



(Credit: Google Image)

Students can watch the online lectures or read suggested study material (before class) and complete problem sets & preparing for quizzes/ exams (after class)[1]

4.5 Ground Rule

- Learning materials such as NPTEL Video on Web service and its application uploaded on the course website.
- Open to suggestions, sensitive feelings and concerns
- Each activity such as online quiz, discussion and group presentation contains separate assessment and evaluation criteria.
- Final score of a team will be calculated as the sum of individual score and team score.
- Plagiarism is not allowed. If found, the student would be given zero to particular assessment.

4.6 Concept teaching/Video lecture

Initially, a lecture on Service-Oriented Principles was delivered to students based on the **ARCS model**. The context or importance of Service-Oriented Principles in industries for achieving the business automation process is addressed that ensure the attention of the students. Then, a concept in relevance to Airline System was explained. A **Class poll activity** was conducted in the mid of the lecture to alert the lack of student's attention. The students were asked to list the key service-oriented principles such as web services, Orchestration, reusability, operations etc in the Airline System that boost the confidence of individual students. Based on this lecture, students should be able to design web services for the airline system based on service-oriented principles. . A **video lecture** was created to describe the orchestration that is one of the core

components of SOP. The recorded video *uploaded on the YouTube channel*. The URL of the recorded video is given below.

<https://youtu.be/fhtJltuDIIY>.

The theme of the concept is explained in Figure 1. An *NPTEL video lecture* on web service and its application was uploaded on Google Drive and shared with students.

URL: <https://drive.google.com/drive/u/0/folders/1q-B4-TpR-qzfQZ6dcIOisELIvkFs1Ee2>.

The other study materials also posted on the course website one week before the commencement of in-class activity.

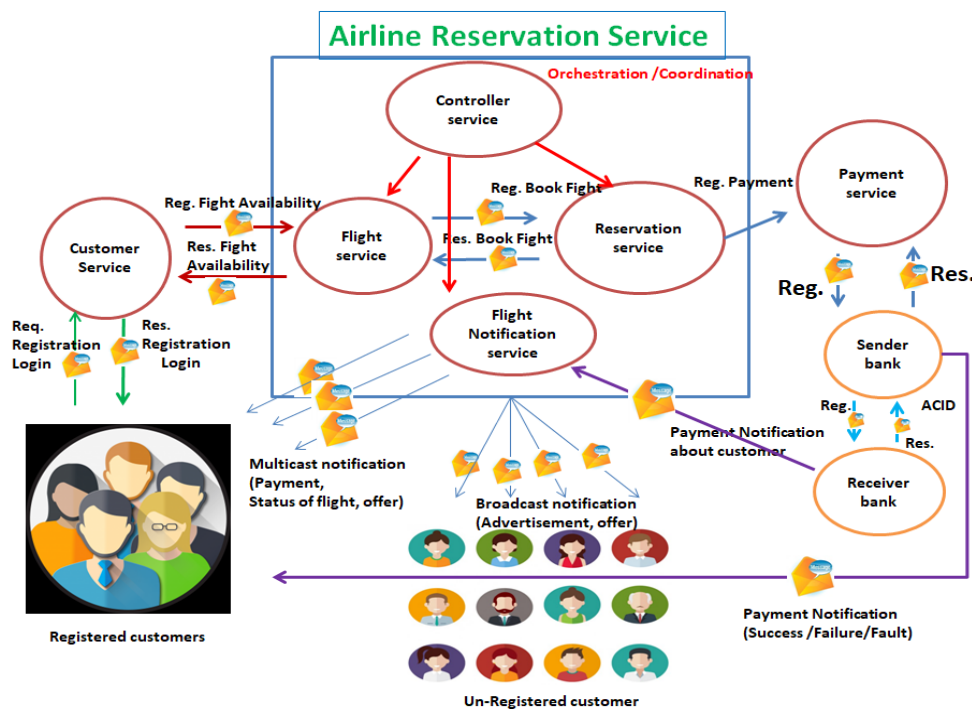


Figure1: Airline Reservation System

4.7. Justification to form a group

A heterogeneous group with six members was formed. Each group formed by bright students, average students, and slow learners. Classification of students such as *a bright student, average students, and slow learners* is based on their academic performance. The total class strength is 50 (Boys and Girls).

Eight heterogeneous groups were formed based on academic performances such as the CGPA of end semester examination and Internal Assessment Test – 2. Table 1 shows the academic performance of the students.

Table1: Academic performance of the students

CGPA up to 7th semester		Internal Assessment Test - 2	
Bright students(CGPA > 8.0)	17	Bright students(CGPA > 8.0)	19
Average students(CGPA > 6.0 <8.0)	25	Average students(CGPA > 6.0 <8.0)	26
Slow learners(CGPA < 6.0)	8	Slow learners(CGPA < 6.0)	5

The CGPA is considered up to the 7th-semester to form the groups. *Six heterogeneous groups* formed with 36 students in which 6 students form a group. The distribution of bright students, average students, and slow learners is given in table 2.

Table 2: Distribution of different individuals

Group 1,2,3,4, 5, 6	
Bright students	02
Average students	03
Slow learners	1

Group 7 consists of 02 bright students, 04 average students and one slow learner. *Group 8* consists of 03 bright students, 03 average students and one slow learner.

4.8 Schedule of flipped class

Week end Friday	Before class At home Holidays	During the class at home			In class 25.10.2019
		Monday	Wednesday	Thursday	
Study material to be posted in the course web site	Watch video	Review material Discussion	Online Quiz 15 minutes	Discussion Board (Educator address the gap)	Brainstorming /Presentation 50 minutes Venue: CSE Seminar Hall

4.9 Flipped class activities and assessments

I conducted three activities in Flipped class collaborative learning that is listed below.

1. Online Discussion
2. Online Quiz
3. Brainstorming/Presentation

Table 1: Formative and summative assessment

Team #	Graded Discussion	Online Quiz	Brainstorming/Group Presentation	
	Individual score (GDIS)	Individual score (QIS)	Group Individual Score (BSIS)	Group Presentation-Group Score (BSGS)
	10 Marks	15 Marks	5 marks	20 Marks
Team N	GDIS1	QIS1	BSIS1	GPS
	GDIS2	QIS2	BSIS2	
	GDIS3	QIS3	BSIS3	
	GDIS4	QIS4	BSIS4	
	GDIS5	QIS5	BSIS5	
	GDIS6	QIS5	BSIS6	
Total (Team _i)	$\frac{\sum_{i=1}^n GDIS_i}{n}$	$\frac{\sum_{i=1}^n QIS_i}{n}$	$\frac{\sum_{i=1}^n BSIS_i}{n}$	GPS

Where , n= 1 to 6 | 7

GDIS- Graded Discussion Individual Score

QIS- Quiz Individual Score

BSIS-Brainstorming Individual Score

GPS- Group Presentation Score

4.9.1 Evaluation of Individual and Group Performance

Students' performances were evaluated using formative and summative assessment techniques. It is an effective tool to evaluate the various skills of the students.

Assessment 1: Discussion

The objectives of the discussion are to encourage *peer learning, critical thinking, and reflection*. I gave a discussion topic on the “Business Automation Process”. Based on the concepts of teaching and reading materials, students able to post their ideas and respond to the post of at least one of their friends. The following evaluation criterion was used for discussion to assess the students' performance.

Graded Discussion: evaluation criteria

Here, I used *IIEECP graded discussion rubrics*. I would like to thank IICEEP.

- Your initial response is within due. This gives a few days to you and your peers to comment on each other's posts.
- You will not be able to see your peers' post till you have posted your own

- The discussion closes on deadline.

Individual Formative assessment: Graded discussion – Individual Score (GDIS_i)

- The initial response is posted within the deadline (GDIS1)- (1 mark)
- The question/all parts of the question are addressed directly(GDIS2)- (3 marks)
- The response is well reflected, and based on personal opinion and experience(GDIS3)- (3 marks)
- Active/constructive interaction with different peers' posts((GDIS4) (2 marks)
- The posts are well composed and proofread ((GDIS5)-(1 mark)

$$GDIS_i = GDIS1 + GDIS2 + GDIS3 + GDIS4 + GDIS5, \text{ where } i = 1 \text{ to } 6 \mid 7$$

Assessment 2: Online Quiz.

It carries 15 marks. It contains 15 multiple choice type questions from the study resources. The objective of an online quiz is to get students understanding level during the activity. The result used to address the gaps.

Individual Formative assessment – Online Quiz (QIS_i)

- Correct Answer (QIS1)
- Response time to take the quiz – Bonus (QBIS_i)

$$QIS_i = QIS1 + QBIS_i, \text{ where } i = 1 \text{ to } 6 \mid 7$$

Assessment 3: Brainstorming (BS)

The goal of the Brainstorming method is used to develop ideas by students. *Students can freely propose their new ideas in their peer group.* Team members form a circle and discuss their ideas. Simultaneously, the performances of the students were assessed during the constructive discussion and were evaluated the solution of individual students using the following evaluation criteria. The students' performance was assessed individually and group with *individual formative assessment and Group summative assessment*, respectively. The group summative assessment follows the analytic rubric for evaluation. Through this, students should be able to develop various skills like *leadership, interpersonal skill and team, and time management skill.*

Individual formative Assessment- Brainstorming (BSIS_i)

- Constructive idea /subject knowledge – 4 marks(**BSIS₁**)
- Individual Role of his/her sub-task- 2 marks (**BSIS₂**)
- Communication skill – 2 marks(**BSIS₃**)
- Body language - 2marks (**BSIS₄**)

$$BSIS_i = BSIS_1 + BSIS_2 + BSIS_3 + BSIS_4 \text{ where } i = 1 \text{ to } 6 \mid 7$$

Group Summative Assessment – Presentation (BSS)

The following rubric used for presentation.

Evaluation criteria	Excellent 5.0 to >3.0 pts	Good 3.0 to >2.0 pts	Satisfaction 2.0 to >1.0 pts	Marks
Objectives	Presented clear objectives.	Presented objectives that somewhat relevant to the concepts.	Presented wrong objectives.	5
Constructive Idea	Excellent idea with evidence and relevant content to demonstrate the problem.	Little relevant content to demonstrate the problem.	No relevant content to demonstrate the problem.	5
Solution	Delivered the solution and content professionally and answered the quires	Delivered the contents and answered the few quires.	Delivered the contents and not answered the quires.	5
Total				15

4.9.2 Final Individual score

The final individual score will be calculated based on the formative and summative assessment method for flipped class assessments such as Graded Discussion, online Quiz and Brainstorming/Presentation. The final score is calculated by a given formula.

$$\text{Final Individual Score (FIS}_i\text{)} = GDIS_i + GIS_i + GPS + BSIS_i$$

4.10. Feedback questionnaires for Flipped classroom

The feedback questionnaires were posted on the Canvas Course website after the completion of all activities. The feedback was collected from the students using the following questionnaires. It is used to assess the success of the Flipped Classroom. The first question is used to assess the *student's willingness* whether the students are like this activity or not. The second and third questions measure the *objectives and outcomes of this activity*. The fourth question asks the students *comments to improve the learning strategies*.

FEEDBACK

1. I like Flipped Classroom learning activity

Yes

No

2. The Flipped classroom gives me greater opportunities to enhance my interpersonal skills like communication, listening, team work and leadership.

Strongly Agree

Agree

Disagree

3. Flipped Classroom will help me in my future education

Yes

No

4. Any other comments

4.11 Feedback Analysis

Attempts: 30 out of 30

+0.56

I like Flipped Classroom learning activity

Discrimination Index
②

Yes	26 respondents	87 %	✓
No	4 respondents	13 %	

87% answered correctly

Attempts: 30 out of 30

+0.49

I like the video lectures. It will useful me when I was absent a lecture.

Discrimination Index
②

10 minutes video	4 respondents	13 %	✓
20 minutes	17 respondents	57 %	
45 minutes	8 respondents	27 %	
Don't like	1 respondents	3 %	

13% answered correctly

Attempts: 30 out of 30

+0.67

Discrimination Index
?

The Flipped classroom gives me greater opportunities to enhance my interpersonal skills like communication, listening, team work and leadership.

Strongly agree	10 respondents	33 %	✓
Agree	20 respondents	67 %	
Disagree		0 %	

33% answered correctly

Attempts: 30 out of 30

+0.41

Discrimination Index
?

Flipped Classroom will help me in my future education

Yes	27 respondents	90 %	✓
No	3 respondents	10 %	

90% answered correctly

Any Other comments:

The students posted the following comments

- *Nothing*
- *Good*
- *Continue doing activities*
- *NO sir*
- *Mm*
- *Flipped Classroom is very useful and i like that class very much*
- *Thank you for your cooperation with us throughout the semester*
- *It was a wonderful experience*
- *over all good*

The above statistics clearly show that **87 % of the students** like the flipped class. All the students gave excellent comments regarding the second questioner. **All the students agreed** that the flipped class enhanced their inter-personal skills such as communication, listening, teamwork and leadership. 90 percent of the students told that the flipped class will be used in their future.

4.12 Reflection critique

Challenges

- Recording video lecture and delivering content in front of the camera is challenging.
- Recording Quality video and getting the quality device.
- Deadlock situation may happen during an in-class activity like brainstorming

Steps to be taken to avoid the problem.

- After recording a video with the help of my colleague, I gave it to peer-review to get both positive and negative suggestions/feedback from my peers. Based on their suggestion, I prepared a quality short video lecture.
- The Presentation (PPT) is integrated with video using effective software like Camtasia studio.
- Tips were given to motivate the students during the participation of in-class activity

Post Implementation

1. Based on the observation, a lot of *incidental learning* happening in the discussion forum. They observed and learned a few things individually that lead to self-learning. They *actively involved* in this session rather than it being a one-way communication.
2. All the groups performed well in the group presentation session. This session *re-enforced their understanding*.
3. Always, all the students took the *quiz with joyfulness*.
4. Slow learners from two groups were not comfortable in their presentation due to lack of attention. However, they understood the concepts by these learning methods.
5. Learning at home automatically becomes *student-centric* rather than teacher-centric

4.12 Relevance of program outcomes

Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Outcome(O)	3					2		2	2	2		2

4.13 Conclusions

The selected topic is playing a vital role in the industry to develop a business automation model. The flipped classroom technique is used to assess the students' performance in continuous progress. This collaborative learning consists of three assessments such as *Graded Discussion, Online Quiz, and Group presentation*. The student's performance was assessed by formative and summative evaluation tools to achieve objectives and outcomes. Based on the feedback analysis of the Flipped classroom learning technique and observation, the students could be able to enrich various skills such as *communication, cooperation, and leadership*.

References:

1. Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™
2. Stephanie butler velegol, sarah e. Zappe, emily mahoney, The Evolution of a Flipped Classroom: Evidence-Based Recommendations, winter 2015, pp.1-37
3. Dian Mita Nurhayati and Hartono, Implementation of Cooperative Learning Model Type with RME Approach to Understanding of Mathematical Concept Student State Junior High School in Pekanbaru, AIP Conference Proceedings 1848, 040002, 2017, <https://doi.org/10.1063/1.4983940>
4. Micheal M van Wyk, The Effects of the Cooperative Learning Method on Student Achievement, Attitude and Motivation in Economics Education, Journal of Social Science pp. 261-270, 2017, <https://doi.org/10.1080/09718923.2012.11893104>